

Deacetylation modification techniques of chitin and chitosan

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5.1 Introduction

Chitin, a linear (1-4)-2-acetamido-2-deoxy- β -D-glucan, is the second most abundant natural polymer next to cellulose (Roberts, 1992). It is the major exoskeleton component of various crustacean species like crab, shrimp, etc. (Mincea et al., 2012). Even though chitin is abundant and easily accessible, its use as a potential biomaterial is limited. This is due to its bulk and highly crystalline structure which makes it insoluble in water and other commonly used solvents such as alcohols, chloroform, etc. (Tsigos et al., 2000). The major derivative of chitin is chitosan, a cationic (1-4)-2-amino-2-deoxy- β -D-glucan. Chitosan has attracted attention for biomedical applications owing to its superior characteristic properties, such as degradability, solubility in weak acids, pH-sensitivity, biocompatibility, and absence of toxicity (Wijesena et al., 2015; Zhou et al., 2014).

During the process of deacetylation of chitin, the acetyl (C_2H_3O) group is removed resulting in the formation of an amino group (NH_2). This process first starts in the amorphous regions of chitin material proceeding toward the crystalline regions (Tahtat, 2007; Zvezdova and Uzov, 2012). The deacetylation process results in a copolymer of *N*-acetyl-glucosamines and glucosamines. When the resulting copolymer contains more than 50% *N*-acetyl-glucosamine units, it is typically referred to as chitin. For chitin polymers, the percentage of *N*-acetyl-glucosamine units is termed degree of acetylation (DA) and can vary from 50% to 100%. On the other hand, when the resulting copolymer contains more than 50% glucosamine units, it is typically referred to as chitosan. For chitosan polymers, the percentage of glucosamine units is termed degree of deacetylation (DDA) and can vary from 50% to 100%. DA and DDA are related so that $DA + DDA = 100\%$. The generation of the highly reactive amino groups in the chitosan polymer imparts versatility in its use for food, textile, cosmetics, environmental, biomedical, and pharmaceutical applications. Due to the presence of amino groups, chitosan is easily soluble in weak acids, a characteristic distinguishing it from chitin, and forms viscous solutions that can be formed into films, sponges, beads, etc.

During deacetylation process, depolymerization of chitin polymer may also occur due to the extreme process conditions (e.g., strong base, high temperatures and pressures). This depolymerization leads to the production of water-soluble low-molecular weight polymers and oligosaccharides as a product of the reaction (Wijesena et al., 2015). Since the DDA affects chitosan physical, chemical, and biological properties, it is necessary to monitor the DDA (Martinou et al., 1995). For example, chitosan

solubility in aqueous environment depends on the relative number of D-glucosamine and N-acetyl-D-glucosamine monomers present (Dimzon and Knepper, 2015). This in turn depends on the partial deacetylation of chitosan where samples with a DDA of 45–55% display good solubility in weak acids due to protonation of the amino group (Ramirez-Coutino et al., 2006). In applications where mechanical properties or water transmission rates are important, chitosan films with higher DDA exhibit improved tensile strength, elongation, and water transmission rates, which is due to the changes in the molecular arrangement in the film state of the chitosan. Presence of increased percentage of amino groups in chitosan films with high DDA favors the formation of inter- and intramolecular hydrogen bonding which in turn results in ordered molecular structure, increased crystallinity, and hence increased mechanical properties (Potivas and Laokuldilok, 2014). Degradation rate is inversely proportional to DDA (Nordtveit et al., 1994) where chitosans with lower DDA degraded faster when compared to chitosans with higher DDA (Zhou et al., 2014). Differences in degradation rate between chitosans with different DDAs are also attributed to crystallinity effects in the polymer. Chitosans with high DDAs have fewer acetyl groups, which reduces steric hindrance and leads to tighter chain packing, increased crystallinity, and thus slower degradation rate (Yuan et al., 2011). Furthermore, it has been reported that chitinases (enzymes that degrade chitin molecules into oligomers) exhibit increased catalytic activity on partially deacetylated chitin than on chitin or chitosans with high DDA (e.g., >95% DDA) due to inefficient chain packing (Ramirez-Coutino et al., 2006). Chitosan membranes with percentage of DDA between 65 and 80 induced inflammatory reactions which subsided with time resulting in degradation of films, osteogenesis, and granulation tissue formation (Yuan et al., 2011). In addition to the above properties, DDA will also influence biocompatibility, biodegradability, hydrophilicity, mucoadhesion, hemostatic, analgesic, anticholesterolemic, antioxidant, and adsorption-enhancing properties of chitosan-based biomaterials (Zajac, 2015; Freier et al., 2005; Aranaz et al., 2009). Therefore DDA may be used as a major factor in matching chitosan to specific industrial or biomedical requirements (Zhou et al., 2014).

In this chapter, different techniques available for deacetylating chitin samples for the preparation of chitosan and its derivatives for various applications are discussed. Additionally, different methods used for the determination of percentage of DDA of chitosan are discussed in detail.

5.2 Deacetylation methods

5.2.1 Alkali treatment method

There are several methods currently used for deacetylation of chitin, of which hot alkali treatment is commercially and extensively used. In this method, sodium hydroxide (NaOH) is commonly used for the deacetylation process (Fig. 5.1). The factors which influence the extent of deacetylation are concentration of NaOH, reaction time, temperature, chitin or chitosan concentration, and molecular weight of the initial polymer (Potivas and Laokuldilok, 2014).

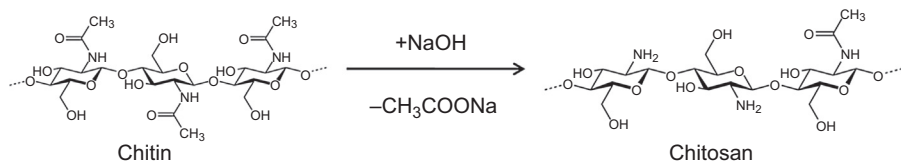


Figure 5.1 Reaction scheme for alkaline deacetylation of chitin.

5.2.1.1 Experimental procedure

1. Chitin (10 g) is mixed with 100 mL of 50% NaOH. Alkali concentration may vary and generally effects rate of deacetylation and polymer degradation.
2. This mixture is stirred with a magnetic stirrer at $100 \pm 1^\circ\text{C}$ for 30–120 min. Reaction times may be varied to obtain different DDA.
3. After the deacetylation process, the product is rinsed several times with hot distilled water.
4. Keep at 80°C for 3.5 h or until completely dry (Wojtasz-Pająk and Szumilewicz, 2009).

In general, reaction times have effect on DDA and polymer stability. About 82% deacetylation is obtained by treatment of chitin with 50 wt% NaOH for 1 h at 100°C . When the reaction time is increased up to 48 h, 100% deacetylation is seen, but there is a decrease in molecular weight of the deacetylated chitin due to chain degradation. Some strategies to minimize chain degradation have been investigated such as reducing the amount of alkali added or by using water miscible solvents like 2-propanol or acetone (Batista and Roberts, 1990). Tahtat et al. (2007) reported that chitin samples pretreated with γ irradiation were more susceptible to *N*-deacetylation by alkali treatment than those that were not irradiated. This may be due to radiation-induced decrease in molecular weight of chitin.

Potivas and Laokuldilok (2014) found that there is a linear relationship between the DDA and deacetylation reaction time. They produced chitosan films with different percentages of DDA by using 50% NaOH at a temperature of about 95°C for different time points between 2 and 5 h. Their results showed that at 120 min of deacetylation, DDA of chitosan was about 15.24%, and the highest amount of DDA was obtained at 300 min (70.19%). It is pointed out that there are differences in reported rates of deacetylation by different groups which are largely attributed to differences in source of original chitin source (e.g., shrimp vs. crab, vs. fungi) (Ocloo et al., 2011; Abdou et al., 2008).

The *N*-deacetylation of chitin can be performed homogeneously or heterogeneously. Chitosan with a more regular distribution of the residual acetamide groups can be obtained from chitin using homogenous deacetylation when compared to heterogeneous method. Homogenous deacetylation of chitin is done in the presence of NaOH under room temperature, which results in the formation of water soluble chitosan with the DDA between 48% and 55%. Whereas in heterogeneous deacetylation, chitin is treated with hot, concentrated solution of NaOH which results in the formation of insoluble chitosan exhibiting DDA of about 85–93% (Tsigos et al., 2000). Even though both of these methods efficiently deacetylate chitin, they suffer from following disadvantages: (1) consumption of high quantities of energy; (2) large amounts of

alkali solution are required; and (3) the products formed exhibit a range of DDA and broad molecular weight distribution (Tsigos et al., 2000; Tan et al., 2015).

5.2.2 Alkali treatment using microwave technique

Microwave techniques have been employed for use in polymer synthesis, chemical modification, and polymerization reactions due to the immediate heating and vibrational energy at the molecular level. Lertwattanaseri et al. (2009) performed microwave-assisted chitin deacetylation using NaOH as an alkaline solvent to increase efficiency.

5.2.2.1 Experimental procedure

1. Three grams of chitin is hydrolyzed in 300 mL of 3N HCl at 105°C for 3 h and then washed thoroughly with deionized water.
2. The product is again treated with 300 mL of 3N HCl at 105°C for 6 h and then dialyzed with distilled water until neutral pH is achieved to obtain chitin whiskers.
3. Chitin whiskers are treated with 40% w/v of aqueous NaOH at 150°C under N₂ atmosphere in a microwave chamber (power, 400 W) at frequency of 2.45 GHz for the periods of 1–6 h.
4. The resulting product is dialyzed with distilled water until neutral pH is achieved. The chitosan material is then dried.

Fourier transform infrared (FTIR) spectroscopy results confirmed that the microwave technique was effective in deacetylating the original chitin to create chitosan materials. The percentage of DDA achieved was more than 90% in 3 h as compared to as long as 21 h in conventional alkali treatment. Authors claimed that the microwave-mediated method helped to carry out the deacetylation process under highly alkaline conditions (more than 40% w/v).

5.2.3 Enzymatic deacetylation

Apart from alkaline deacetylation, enzymatic deacetylation using chitin deacetylases obtained from different biological sources such as fungi (Martinou et al., 1995; Tsigos et al., 2000) and insects (Tsigos et al., 2000) presents an efficient alternative for the production of chitosan derivatives due to its controllable activity and minimization of depolymerization (Martinou et al., 1995; Tsigos et al., 2000). Chitin deacetylation by enzymatic processes results in the production of a homogenous product having more block defined pattern of deacetylation. Moreover, this method is more ecofriendly than the thermochemical process. However, this method is expensive due to the use of high-cost chitin deacetylases (Kashyap and Garg, 2014).

Chitin deacetylases (EC 3.5.1.41) are part of the hydrolase family of enzymes, as they cleave C—N bonds other than peptide bonds in linear polyamides. They are also known as chitin amidohydrolases as they catalyze amino sugar metabolism (Kashyap and Garg, 2014). These enzymes were first isolated from the mycelium of the fungus *Mucor rouxii* (Zhao et al., 2011), and other fungal species which produce them include *Absidia coerulea*, *Aspergillus nidulans*, and two strains of *Colletotrichum lindemuthianum* (Tsigos et al., 2000; Ghormade et al., 2010). These enzymes are glycoproteins

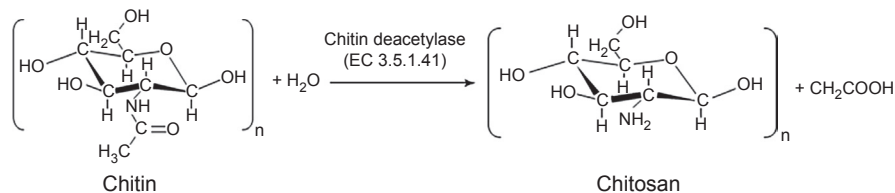


Figure 5.2 Enzymatic deacetylation of chitin.

with high thermal stability, narrow specificity toward water soluble chitin and its derivatives (Tsigos et al., 2000; Martinou et al., 1995). The chitin deacetylases from the fungal species of *A. nidulans* and *C. lindemuthianum* are not inhibited by the product of deacetylation, i.e., acetate (Tsigos et al., 2000).

The chitin deacetylases catalyze the hydrolysis of *N*-acetamido bonds of chitin thereby converting it into chitosan as shown in Fig. 5.2 (Tsigos et al., 2000; Ghormade et al., 2010; Kashyap and Garg, 2014). A chitin oligosaccharide deacetylase from *Vibrio cholerae* specifically deacetylates the second unit from the nonreducing end of chitin oligomers, leading to the formation of monodeacetylated chitosan oligomer (Hamer et al., 2014). Martinou et al. (1995) defined one unit of chitin deacetylase activity as the quantity of enzyme required to generate 1 mol of acetate/minute when incubated with hexa-*N*-acetyl chitohexose. They also standardized the assay conditions for chitin deacetylase activity as 15 min incubation at 50°C to achieve optimum sensitivity and speed (Tsigos et al., 2000).

Chitin deacetylases from *M. rouxii* effectively deacetylated all the *N*-acetyl-D-glucosamine units present in both chitin and chitosan samples. But when crystalline and amorphous chitin samples were incubated with the enzyme for 20 h, about 0.5% and 95% deacetylation was obtained, respectively. This infers that this enzyme is not efficient in deacetylating insoluble chitin, likely due to the decreased diffusion of aqueous enzyme solutions into highly crystalline polymer; therefore pretreatment of chitin to convert it from crystalline to amorphous state is required. The factors that influence enzymatic deacetylation of insoluble chitin are the structure of chitin, properties of chitin deacetylase enzyme, surface area exposed to enzyme, and the mode of interaction between enzyme and chitin molecules (Martinou et al., 1995).

5.2.4 Steam explosion method

Another method of deacetylating chitosan is based on steam explosion (Tan et al., 2015). It is an effective and green hydrothermal treatment method. Here the chitin is treated in a puffing gun (Fig. 5.3) with saturating steam at increased pressure and temperature for several minutes, which is followed by explosive decomposition. The steam applied to the biomass is converted to thermomechanical force under high pressure, which destroys the chitin. The intermolecular interactions of molecules are broken by steam explosion which leads to the formation of free polymer chains. Steam explosion represents an alternative method for chitosan production. In this method, chitin with 75% moisture content exhibited maximum DDA (43.7%) when compared to chitin containing 50% and 35% of moisture which resulted in only 40% and 32%

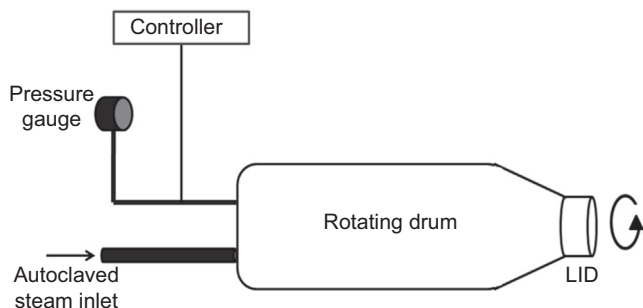


Figure 5.3 Schematic representation of puffing gun used for steam explosion.

DDA, respectively. Since hydroxyl groups react with *N*-acetyl groups, chitin samples may undergo greater DDA in the presence of high moisture content (Tan et al., 2015).

5.3 Methods for determination of degree of deacetylation

To utilize the chitosan samples for various industrial and biomedical applications, it is essential to determine the DDA as this parameter affects the property of the chitosan samples. Different methods are employed for the determination of DDA of chitosan. Of these, proton nuclear magnetic resonance spectroscopy, (^1H -NMR), potentiometric titration and FTIR are frequently used (Dimzon and Knepper, 2015).

5.3.1 Fourier transform infrared spectroscopy (FTIR)

FTIR is a widely used technique for the determination of DDA of chitosan and uses two absorption bands: a characteristic band (probe band) that represents the amide group of *N*-acetyl-D-glucosamine residues and a reference band which represents a group that is present in both the monomers (D-glucosamine and *N*-acetyl-D-glucosamine) (Dimzon and Knepper, 2015). The most commonly used probe and reference bands are the peaks at 1652 and 3450 cm^{-1} which correspond to the amide I and hydroxyl groups of chitosan, respectively (Fig. 5.4) (Wijesena et al., 2015; Abdulkarim et al., 2013; Tan et al., 2015; Baskar and Sampath Kumar, 2009; Hussain, 2013). Using these absorption bands, the percentage of DDA is calculated as follows,

$$\text{DDA \%} = 100 - \frac{A_{1652}}{A_{3450}} \times 115$$

The above equation clearly shows that lower the absorption of the amide I group, higher the DDA of chitosan material, and this absorbance is found to decrease by increasing the time of deacetylation process (Hussain, 2013). Apart from this calibration equation, many researchers have utilized different peaks and proposed various

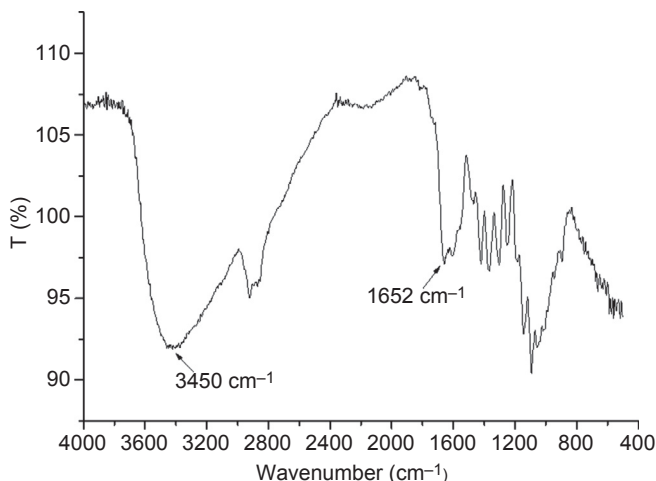


Figure 5.4 Typical fourier transform infrared spectra of chitosan.

calibration curves for the determination of percentage of DDA (Tahtat, 2007; Baxter et al., 1992). The FTIR method for determining DDA has several of advantages as well as disadvantages. Advantages are that the reaction is relatively fast, uses cheap chemicals, uses widely available equipment, and does not require purity of the sample or dissolution of the sample in aqueous solvent (Baxter et al., 1992). The major disadvantage of this method is that even a small variation in the peak intensity, due to instrument and baseline settings, will have a great affect on calculated DDA values. Because of this, FTIR is typically used only for rough estimation of DDA, and care must be taken while using this method. Other drawbacks are that absorbance of some of the peaks may be influenced by humidity and peaks resulting from impurities may overlap with the signals from chitosan (Czechowska-Biskup, 2012).

5.3.2 ¹H-NMR spectroscopy

¹H-NMR spectroscopy is the most successful method for the determination of percentage of DDA. In this method, a known quantity of chitosan sample is dissolved in deuterated aqueous solutions of DCl/D₂O, and the spectra between 0 and 6 ppm are measured (Fig. 5.5) (Czechowska-Biskup, 2012). The integrals (I) of peak areas of the *N*-acetyl group at 2.05 ppm (CH₃) and at 3.15 ppm (H₂) are considered in the following equation which is used to calculate percentage of DDA (Lertwattanaseri, 2009).

$$\text{DDA } \% = \left\{ 1 - \left(\frac{I_{\text{CH}_3/3}}{I_{\text{H}_2}} \right) \right\} \times 100$$

The DDA may also be calculated using various proton signals (Table 5.1). For example, the integrals of the peak of proton H1 of deacetylated monomer (H1-D) and

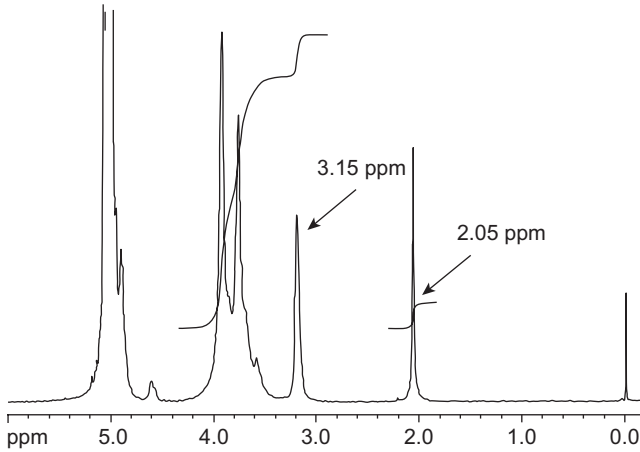


Figure 5.5 ^1H -NMR spectra of chitosan.

Table 5.1 Chemical shifts for proton in $\text{CD}_3\text{COOD}/\text{D}_2\text{O}$ or $\text{DCI}/\text{D}_2\text{O}$ solution at 65°C (Kasaai, 2010; Vårum et al., 1991; Hirai et al., 1991)

Type of proton	Position (δ , ppm)
H_1 (H_1 of GluNAc or $\text{H}-1(\text{A})$)	4.62–4.85
H_1 (H_1 of GluNH $_2$ or $\text{H}-1(\text{D})$)	4.85–4.97
H_2 (H_2 of GluNH $_2$ or $\text{H}-2(\text{D})$)	3.18–3.24
H_2 (H_2 of GluNAc or $\text{H}-2(\text{A})$)	3.38–3.65
H_3 (H_3 of GluNH $_2$ or $\text{H}-3(\text{D})$)	3.52–3.87
H_3 (H_3 of GluNAc or $\text{H}-3(\text{A})$)	3.52–3.65
$\text{H}_3, \text{H}_4, \text{H}_5, \text{H}_6, \text{H}_6'$	3.74–4.34
$\text{HN}-\text{COCH}_3$ or $\text{H}-\text{Ac}$	1.95–2.09

of the peak of the three protons of acetyl group ($\text{H}-\text{Ac}$) are considered to calculate percentage of DDA as follows (Kasaai, 2010; Lavertu et al., 2003).

$$\text{DDA \%} = \left(\frac{\text{H1-D}}{\text{H1-D} + \text{H-Ac}/3} \right) \times 100$$

Using the signal from protons $\text{H}_2, \text{H}_3, \text{H}_4, \text{H}_5, \text{H}_6, \text{H}_6'$ (H_2 -6) of both the monomers and the peak of acetyl group ($\text{H}-\text{Ac}$), percentage of DDA is calculated as follows.

$$\text{DDA \%} = \left(1 - \left(\frac{\frac{\text{H-Ac}}{3}}{\frac{\text{H}_2-6}{6}} \right) \right) \times 100$$

The percentage of DDA is calculated by using the peaks of protons H1 of both deacetylated and acetylated monomer (H1-D, H1-A) for DDA lower than 90% using the following equation.

$$\text{DDA \%} = \left(\frac{\text{H1-D}}{\text{H1-D} + \text{H1-A}} \right) \times 100$$

Czechowska-Biskup et al. (2012) in their experiment comparing different methods to determine the DDA confirmed that NMR is the most reliable, rapid, simple, and accurate technique when compared to other methods. However, use of this method may be disadvantageous due to the expense of the apparatus and chemicals used, requirement of trained staff to use deuterated solvents, and limited solubility of chitosan.

5.3.3 UV spectrophotometric analysis

DDA of chitosan can also be calculated by employing UV spectrophotometric analysis since the *N*-acetyl glucosamine absorbs at 199–203 nm in the UV range, and its absorbance readings are linearly dependent on concentration and are not influenced by the presence of solvent. According to this method, first a standard solution is prepared using *N*-acetyl glucosamine in 0.001 mol/L HCl by varying its concentration (0.01, 0.02, 0.03, 0.04, and 0.05 mg/mL) (Jiang, 2001; Muzzarelli and Rocchetti, 1985). The absorbance of the standard solutions is determined at 199 nm using HCl as reference. Thereafter, a known quantity of chitosan is dissolved in HCl solution, and absorbance is measured at 199 nm. Based on the standard curve, the concentration of the acetyl group in chitosan can be calculated using the formula, $\text{DDA} = 100\% - C_1/C$, where C_1 is the acetyl concentration of sample, and C is concentration of sample.

Wu and Zivanovic (2008) proposed another method to determine percentage of DDA using the first derivative UV method as shown in Fig. 5.6. In this method, the percentage of DDA is calculated as follows.

$$\text{DDA \%} = 100 - \left(\frac{\frac{m_1}{203.21}}{\frac{m_1}{203.21} + \frac{m_2}{161.17}} \right) \times 100$$

where, m_1 is mass of acetyl glucosamine in 1 mL chitin/chitosan solution, calculated from the calibration curve by the corresponding H_{203} (first derivative UV value); m_2 is the mass of glucosamine in 1 mL chitin/chitosan solution, calculated as $m_2 = M - m_1$. The mass of chitin/chitosan (M) in the 1 mL solution was calculated as: $M = (M_1 \times M_3) / (M_1 + M_2)$, where M_1 is mass of solid chitin/chitosan sample taken for analysis (100 ± 10 mg); M_2 is mass of 20 mL 85% phosphoric acid; and M_3 is mass of 1 mL chitin/chitosan solution in concentrated phosphoric acid. Both UV spectrophotometric and first derivative UV methods present a simple and rapid measurement of the percentage of DDA. However, these methods suffer from less accuracy when compared to other methods.

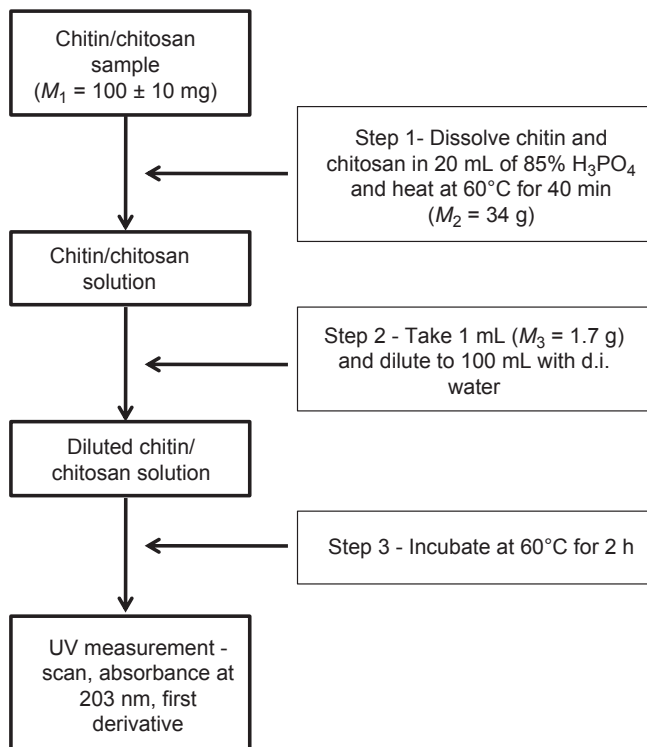


Figure 5.6 Procedure to determine the percentage of degree of deacetylation using 1st derivative UV spectrophotometry method (Wu and Zivanovic, 2008).

5.3.4 Potentiometric titration

Potentiometric titration is considered to be the most consistent and robust technique among the non-NMR methods (Czechowska-Biskup, 2012). However, some of the disadvantages include the time-consuming and labor-intensive nature of measurements and the requirement of solutions of accurately known concentrations (Dimzon and Knepper, 2015). In potentiometric titration, a known quantity of chitosan sample is dissolved in known volume of standard solution of HCl and titrated against a known volume of standard NaOH solution (Broussignac, 1970). The titration curve of pH versus NaOH titration volume is generated and the curve's inflection points are found for each indicated transition (Fig. 5.7). The volume of NaOH consumed at each inflection point is applied to the equation below to determine the amine content of sample;

$$\text{NH}_2 \% = 16.1 \times (Y - X) / M$$

where, M (g) is the weight of chitosan used, X (mL) is the first inflection point on the graph of measured pH versus titration volume, and Y (mL) is the second inflection

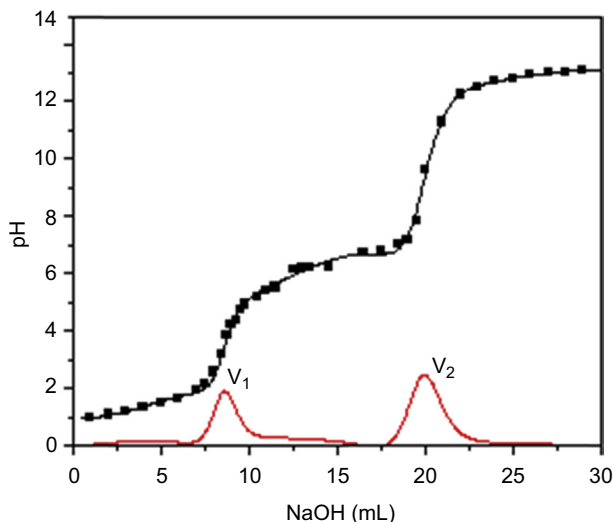


Figure 5.7 A typical potentiometric titration curve.

point. DDA is then calculated based on the theoretical $\text{NH}_2\%$ content (9.94%) using the following equation

$$\text{DDA \%} = (\text{NH}_2 \% / 9.94 \%) \times 100$$

In another titration method, the percentage of DDA can also be calculated using the following formula (Muzzarelli, 1977).

$$\text{DDA \%} = 100 - \left[2.03 \left(\frac{(V_2 - V_1)}{m + 0.0042 (V_2 - V_1)} \right) \right]$$

where, m (g) is weight of sample. V_1 , V_2 (mL) are the volumes of 0.1 M NaOH solution corresponding to the deflection points. It is noted that DDA is calculated directly using this equation—whereas in previous calculation based on volumes of solution at inflection points, the free amine content is calculated and then used in calculation of DDA based on theoretical free amine.

5.3.5 Acid–base titration

Acid–base titration provides results that are comparable to FTIR and ^1H -NMR results. Acid–base titration provides advantages over potentiometric and spectroscopy methods as it is easy, convenient, reproducible, and cost effective (Yuan et al., 2011).

5.3.5.1 Experimental procedure

1. 0.3–0.5 g of chitosan is dissolved in 30 mL of 0.1 M HCl at $20 \pm 5^\circ\text{C}$ with a few drops of methyl orange indicator.

2. 0.1 M NaOH solution is used to titrate the chitosan solution. At the titration point, color changes from pink to orange yellow.
3. To measure moisture content, a known weight (W_1) of chitosan is heated at 105°C until a constant weight (W_2) is reached. Using W_2 and W_1 , moisture content percentage of chitosan is calculated using the formula, $[(W_1 - W_2)/W_1] \times 100$.

The percentage of free NH_2 group can be calculated using the following formula,

$$\text{NH}_2 \% = [(C_1 V_1 - C_2 V_2) \times 0.016] / [G (100 - W)] \times 100$$

where, C_1 is concentration of HCl (M); C_2 is concentration of NaOH (M); V_1 is the volume of HCl added (mL); V_2 is the volume of NaOH added by titration (mL); G is sample weight (g); W is sample moisture content (%); and 0.016 is the theoretical NH_2 content of chitosan (g) in 1 mL of 1 M HCl.

5.3.6 Elemental analysis

The DDA of chitosan may also be calculated by elemental analysis of carbon and nitrogen using an elemental analyzer. The carbon to nitrogen ratio (C/N) varies from 5.145 in completely deacetylated chitosan to 6.861 in chitin, the completely *N*-acetylated polymer. In this approach, DDA can be calculated based on C/N ratio using the following formula, (Abdulkarim et al., 2013).

$$\text{DDA \%} = \left(1 - \frac{\text{C/N} - 5.145}{6.861 - 5.145} \right) \times 100$$

It was found that this method is faster when compared to FTIR analysis (Baskar and Sampath Kumar, 2009). Hussain et al. (2013) used weight percent of carbon (C) or nitrogen (N) from elemental analysis for the determination of percentage of DDA by applying following formulas:

$$\text{DDA \%} = 9600 / (364 \times W_C + 2400) \times 100$$

$$\text{DDA \%} = 1400 / (364 \times W_N) \times 100$$

where W_C and W_N are the weight percent of carbon and nitrogen in the chitosan samples.

5.3.7 Capillary zone electrophoresis method

Wu et al. (2014) developed a method for the determination of DDA using capillary zone electrophoresis (Fig. 5.8). In this method, the electrophoretic mobilities of the chitosan samples are calculated under optimized capillary zone electrophoresis conditions. A calibration curve for the capillary zone electrophoresis method is established using standards with DDA determined through $^1\text{H-NMR}$ ranging from 55.3% to 96.2%. By utilizing this calibration curve, percentage of DDA of the chitosan peaks in the electropherograms was calculated. The data obtained showed very good

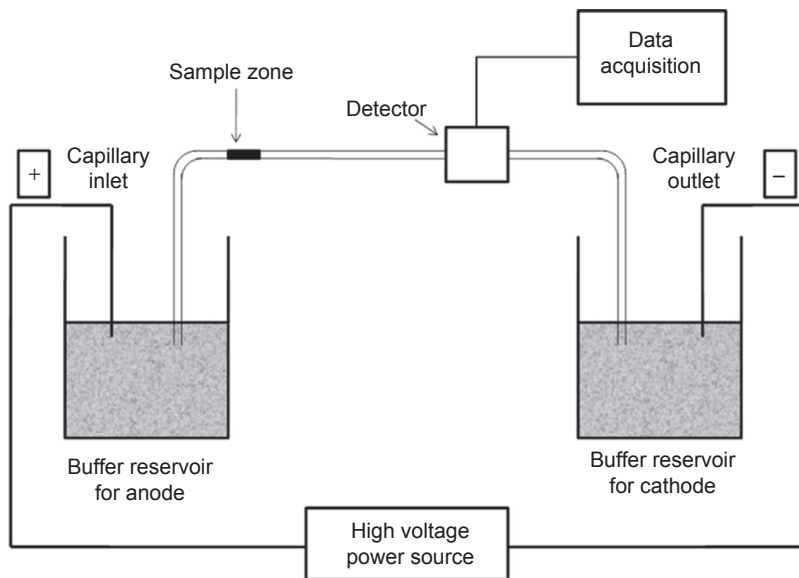


Figure 5.8 Schematic diagram of capillary electrophoresis setup.

agreement with the $^1\text{H-NMR}$ results of the samples. Due to its high resolving power, capillary zone electrophoresis is able to provide separate percentage of DDA of mixed chitosan samples which $^1\text{H-NMR}$ failed to provide. Therefore, this method is highly advantageous for the determination of samples with a wide range of DDA within a given chitosan polymer sample.

5.3.8 Raman spectroscopy

Zajac et al. (2015) reported that Raman spectroscopy is an alternative to FTIR for the determination of DDA with advantages over FTIR, as IR can be used only for rough determination of chitosan DDA. This technique is fast and does not require sample purification or sample dissolution in any solvent. They used FTIR and Fourier transform-Raman spectra to calculate and compare the DDA for six chitosan samples of different DDA. When comparing the FT-Raman spectra of those six samples, there was an intensity change in a few bands with the change in %DDA. The bands at 1654, 1591, and 936 cm^{-1} correspond to $\nu(\text{C}=\text{O})$, $\delta(\text{NH}_2)$, and $\nu(\text{CN})$, respectively, which are related to partial acetylation of NH_2 groups in the chitosan polymer (Fig. 5.9). The following formula describes the relation between the integral intensities and the DDA% of the chitosan samples.

$$A_{1654} = -0.021\text{DDA} + 2.68 \quad (R^2 = 0.9468) \quad \text{for the band at } 1654 \text{ cm}^{-1}$$

$$A_{1591} = 0.016\text{DDA} - 0.92 \quad (R^2 = 0.9787) \quad \text{for the band at } 1591 \text{ cm}^{-1}$$

$$A_{936} = -0.075\text{DDA} + 0.71 \quad (R^2 = 0.9848) \quad \text{for the band at } 936 \text{ cm}^{-1}$$

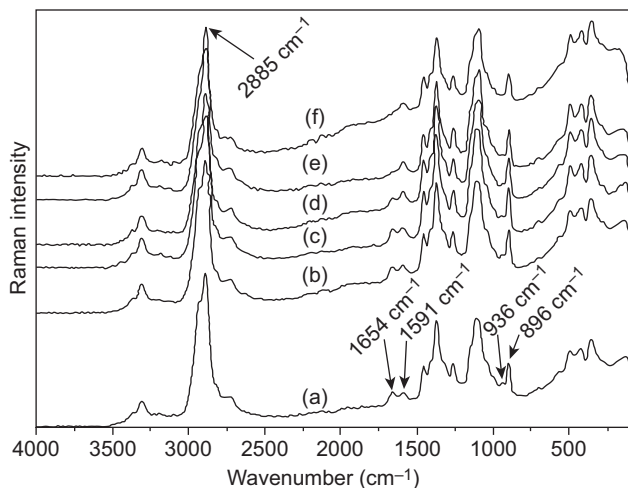


Figure 5.9 Raman spectra of chitosan with DDA = 70% (a); 75% (b); 80% (c); 85% (d); 90% (e); and 95% (f) (Zajac et al., 2015).

The experimental data obtained from FT-Raman analysis exhibited good correlation with FTIR results. However, this method is still in initial stages, and considerable work needs to be carried out to verify the accuracy and reproducibility of the results.

5.4 Conclusions

In recent years, chitin and chitosan biopolymers are receiving increasing attention since they may be widely used in many areas such as biomedical engineering, fiber and textile production, environmental protection, functional food additives, and cosmetics. The quality and chemical properties of both chitin and chitosan are highly dependent on the DDA. In this chapter, we discussed various methods available for deacetylating chitin for the preparation of chitosan for various applications. The alkali treatment method which is widely used suffers from certain disadvantages such as different DDAs, and it also affects the environment. These disadvantages can be overcome by enzymatic deacetylation which is capable of producing chitosan of homogenous deacetylation. Among the various methods employed for determining percentage of DDA, ^1H -NMR is found to be highly reliable and accurate despite of being expensive and time consuming. IR, UV, and titration methods are relatively simple and quick but can be subject to instrument or operator error. Capillary electrophoresis and Raman spectroscopy are emerging technologies that may provide good alternatives for DDA determination. Thus researchers have access to several different methods for determining the DDA of chitosan to standardize its application.

Acknowledgment

This work was supported by DST-Nano Mission (SR/NM/NS-1260/2013), Department of Science and Technology, Government of India.

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